

Open Source Software You Can Use for IoT Development

The Internet of Things (IoT) is spread across various layers, with different types of software being used across them. This article presents a list of open source software packages which can be used in each of the layers.



You might have read Gartner and many other business forecasts on connected devices and IoT growth in the coming years. Billions of connected devices around us, as well as open hardware and software solutions, play a key role in enabling a smarter life in the near future. Before proceeding to the list of open source software in this field, let's have a look at a few layers of the IoT architecture.

- Layer 1 consists of end devices talking to the environment, which are also known as sensor nodes typically powered by microcontroller based targets, wearables and, sometimes, Linux boards.
- Layer 2 consists of gateway devices, typically Linux powered target boards or handheld gadgets for interfacing end devices with cloud platforms and bridging protocols with their middleware support. High end microcontrollers with suitable frameworks can also be a choice here.
- Layer 3 consists of servers, typically deployed on cloud platforms. They provide services like authentication, data collection, data processing, analysis, storage, visualisation, web integration, mobile app connectivity, etc. This may be further divided into various sub-layers or components.

Let's examine some open source choices for development in each layer.

Layer 1

Toolchains: GNU ARM Embedded Toolchain provides

pre-built tool chains as well as source code for Cortex-M and Cortex-R processors, especially for bare metal code for microcontrollers. It has consistent releases in every quarter. The latest stable release is 2016 Q4, which ships with gcc v6.2 and is migrated to 64-bit Linux hosts, whereas the previous release of 2016 Q3 ships with gcc v5.4 and is available for 32-bit Linux. The GNU ARM Eclipse project provides Eclipse CDT extensions and support for popular targets like STM32 F series boards, and FRDM Kinetis KL series boards with C/C++ templates, using this toolchain as the backend. Most of these elements are licensed under GPL.

URL: developer.arm.com/open-source/gnu-toolchain/gnu-rm,gnuarmembedded.github.io

mbed OS: This is specifically designed for IoT needs with common C++ APIs for Cortex-M architecture from various families, and is licensed under Apache 2.0. It comes with an online IDE as well as offline components for development with support for various toolchains, IDEs (like Keil uVision, Eclipse, ARM GCC, IAR, etc) and rich libraries. It accelerates IoT development with good connectivity solutions for Ethernet, Wi-Fi, Bluetooth LE, Thread, 6LoWPAN, LoRaWAN, Cellular, NFC, etc.

URL: developer.mbed.org

Zephyr OS: This Linux Foundation Project is aimed at being a small, scalable, real-time OS for connected and resource-constrained devices, licensed under Apache 2.0. The